- **Data Understanding and Preparation:**

A) Describe the dataset

```
#   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   str
 1   gender            7043 non-null   str
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   str
 4   Dependents        7043 non-null   str
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   str
 7   MultipleLines     7043 non-null   str
 8   InternetService   7043 non-null   str
 9   OnlineSecurity    7043 non-null   str
 10  OnlineBackup      7043 non-null   str
 11  DeviceProtection  7043 non-null   str
 12  TechSupport       7043 non-null   str
 13  StreamingTV       7043 non-null   str
 14  StreamingMovies   7043 non-null   str
 15  Contract          7043 non-null   str
 16  PaperlessBilling  7043 non-null   str
 17  PaymentMethod     7043 non-null   str
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   str
 20  Churn             7043 non-null   str
```

prediction task:
Building a binary classification neural network to predict whether the customer churns or not.

B) Perform necessary data cleaning and preprocessing:

- **Dropping unnecessary columns**
  We drop “customerID” as it has no requirement, neither in EDA nor in model building
- **Data type crosscheck:**
  Column “Total charges” shows a string data type –
  This may happen because of the presence of some string tuple in this feature

```
df["TotalCharges"] = df["TotalCharges"].replace(" ", np.nan).astype(float)
✓ 0.0s
```

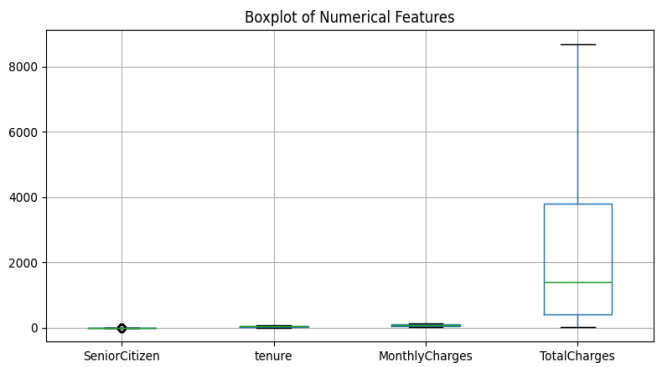
- **Filling missing tuples with the median of the TotalCharges column using –**

```
df['TotalCharges'].fillna(df['TotalCharges'].median())
```

- **Finding duplicate values in the dataset using —-**

```
df.duplicated().sum()
```

- **Checking outliers in numerical features:**

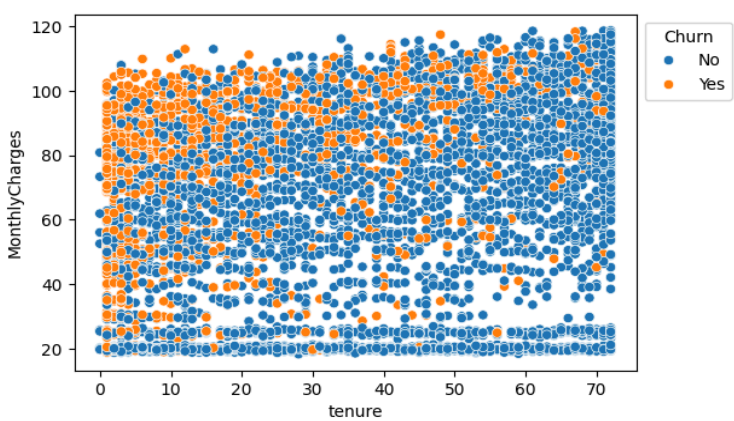


C) Justify feature selection and transformations:

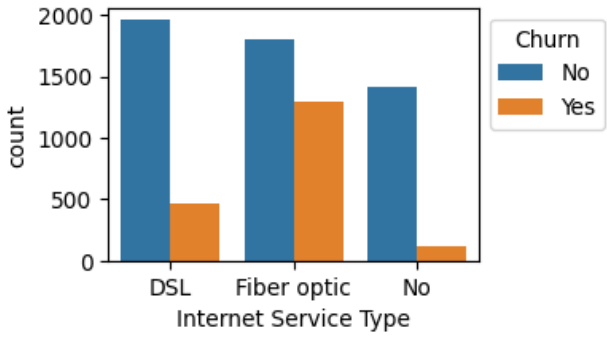
Here, we are just dropping the customer ID only; further feature selection is done ahead

- Exploratory Data Analysis (EDA):

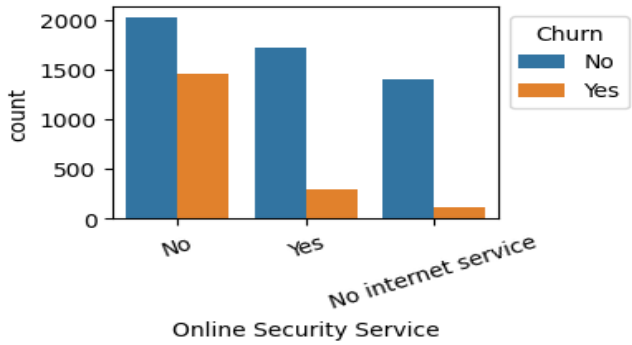
a) Customers who have been using the telco service for longer years are the least prone to churn. The amount of Monthly charges is almost similar for newer and older customers.



b) Customers who use Fiber Optic have the highest propensity to churn, followed by the customers who use DSL services. Customers, who are the least prone to churn, are those who use no internet service.



c) Customers using telco service without internet security churn more than the customers who use internet security. Whereas, customers who do not have an internet connection remain loyal to the telco services.



Usefulness of the insights:

- To analyse the trends of churn among customers
- Reforming the services to prevent further churn.
- Preventing churning customers by giving them offers and discounts.
- Doing product analysis of the competitor company so that similar services can be provided to the customer, and churn can be reduced.

- ## Neural Network Modelling:

1) Build simple feedforward neural networks.

With the top 20 features:
20 neurons in the 1st hidden layer, 10 neurons in the 2nd hidden layer, and 1 neuron in the output layer.

With the top 10 features:
10 neurons in the 1st hidden layer, 5 neurons in the 2nd hidden layer, and 1 neuron in the output layer.

Activation function?
ReLU in the hidden layers, and **Sigmoid** in the output layer.

Why choose 20 and 10 input features?
To reduce **time-consuming** and **computational costs**.

Method to select top features?
Sequential Forward Selection dimensionality reduction technique,

Loss function?
Because of binary classification, we use the **Binary Cross-Entropy loss function** to compute the loss of the output value.

Optimizer?
I have used the **Adam optimizer.** It smooths the **gradient descent** process and avoids getting stuck in plain and steep valleys.

Evaluation matrix?
We will use a confusion matrix to evaluate the model's performance. This matrix will show accuracy, precision, recall, and F1 score.

- ## Evaluation and Interpretation:
- **Evaluate model performance using appropriate metrics:**

```
==================================================
```

For the top **20** features —--

```
training_accuracy: 0.7928647497337593
validation_accuracy: 0.8041163946061036
==================================================
              precision    recall  f1-score   support

         0.0       0.83      0.92      0.87      1036
         1.0       0.69      0.47      0.56       373

    accuracy                           0.80      1409
   macro avg       0.76      0.70      0.72      1409
weighted avg       0.79      0.80      0.79      1409
==================================================
```

For the top **10** features **—**

```
training_accuracy: 0.7930422435214768
validation_accuracy: 0.8048261178140526
==================================================
              precision    recall  f1-score   support
```

```
           0.0       0.84      0.91      0.87      1036
           1.0       0.67      0.51      0.58       373

      accuracy                           0.80      1409
     macro avg       0.76      0.71      0.73      1409
  weighted avg       0.79      0.80      0.80      1409
==================================================
```

- **Discuss what the model captures well and where it may fail:**
  1) Overall accuracy in both models is high.
  2) Both models have predicted non-churn events correctly.
  3) Whereas both models predict the churn events at a low rate
  4) However, overall values of the confusion matrix have improved in the model trained with the top 10 features, with comparison of model trained with the top 20 features. This indicates that eliminating the less relevant features can improve the performance of the model by reducing the effect of noise on the model.

- ## From Insight to Decision:

Decision 1: [Using model's insights]
Run a campaign to retain the high-risk customers.
Action: Identify high churn probability customers and offer incentives.
Assumption: Most customers are price sensitive.
Risk: Revenue loss to the company
Limitations: The model is not trained on social indicators like poverty, accessibility, etc. Thus, depending solely upon incentives may not work well.

Decision 2: [Using EDA insights]
Improve service experience with key pain points.
Action: Improve the optic fibre service; provide online security, device protection, and tech support services at a cheap rate. Provide a yearly contract at a concessional price; improve the electronic check payment method.
Assumption: Most customers prefer quality of service to cost.
Risk: High implementation cost and misinterpretation of the analysis.
Limitations: same as above, the dataset may not cover certain social indicators.